

Cryogenic Evolution of Radiation-Induced Defects in Silicon

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This note extends the project to low temperature operation. The central question is not whether radiation-induced defects disappear as temperature is lowered, but rather how their kinetics and electrical activity change as the system moves from room temperature toward the cryogenic limit.

Main idea

Radiation first creates vacancies, interstitials, and defect complexes. After creation, the subsequent behavior of those defects depends strongly on temperature. As temperature decreases, many thermally activated processes slow dramatically. In particular:

- defect migration becomes strongly suppressed,
- thermal emission from traps becomes slower,
- annealing and recovery pathways freeze out.

Thus, cooling can make the defect landscape more static and more persistent on experimental timescales.

Defect migration at low temperature

A simple first-order model for defect diffusivity is

$$D_{\text{def}}(T) = D_0 \exp\left(-\frac{E_m}{k_B T}\right),$$

where D_0 is a prefactor, E_m is the migration barrier, and k_B is Boltzmann's constant. As T decreases, the exponential factor becomes very small, which implies that vacancies and interstitials become effectively immobile.

Trap emission at low temperature

Many defect states exchange carriers with the conduction or valence band through thermally activated processes. A simple emission-rate model is

$$e(T) = e_0 \exp\left(-\frac{E_a}{k_B T}\right).$$

Equivalently, the emission time constant can be written as

$$\tau_{\text{emit}}(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right).$$

As temperature decreases, trapped carriers may remain captured for much longer times, which can produce persistent charge trapping and history-dependent behavior.

Annealing at low temperature

Many recovery mechanisms are also thermally activated. A simple annealing-rate law is

$$R_{\text{ann}}(T) = R_0 \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right).$$

This implies that as the device is cooled, self-healing and defect reconfiguration become increasingly ineffective.

Physical interpretation near the low-temperature limit

As $T \rightarrow 0$, the important point is not that defects vanish, but that thermal activation becomes negligible. In that limit:

- defect migration tends to stop,
- thermally activated detrapping tends to stop,
- annealing tends to stop,
- static defect potentials and non-thermal processes may still remain relevant.

Therefore, cryogenic operation changes radiation-response physics rather than simply removing radiation damage.

Engineering implication

Room-temperature intuition cannot be applied blindly to cryogenic silicon devices. Low temperature can suppress some noise mechanisms, but it can also lock in radiation-induced defect configurations and extend trap lifetimes. This is important for cryogenic readout electronics, detectors, and quantum-adjacent silicon hardware.